SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	**T Bhanu Teja**	**Reg. No.:**	**192472259**
Section / Batch:	2024 – CSE(AI)	**Date:**	15.10.26

Code of Conduct

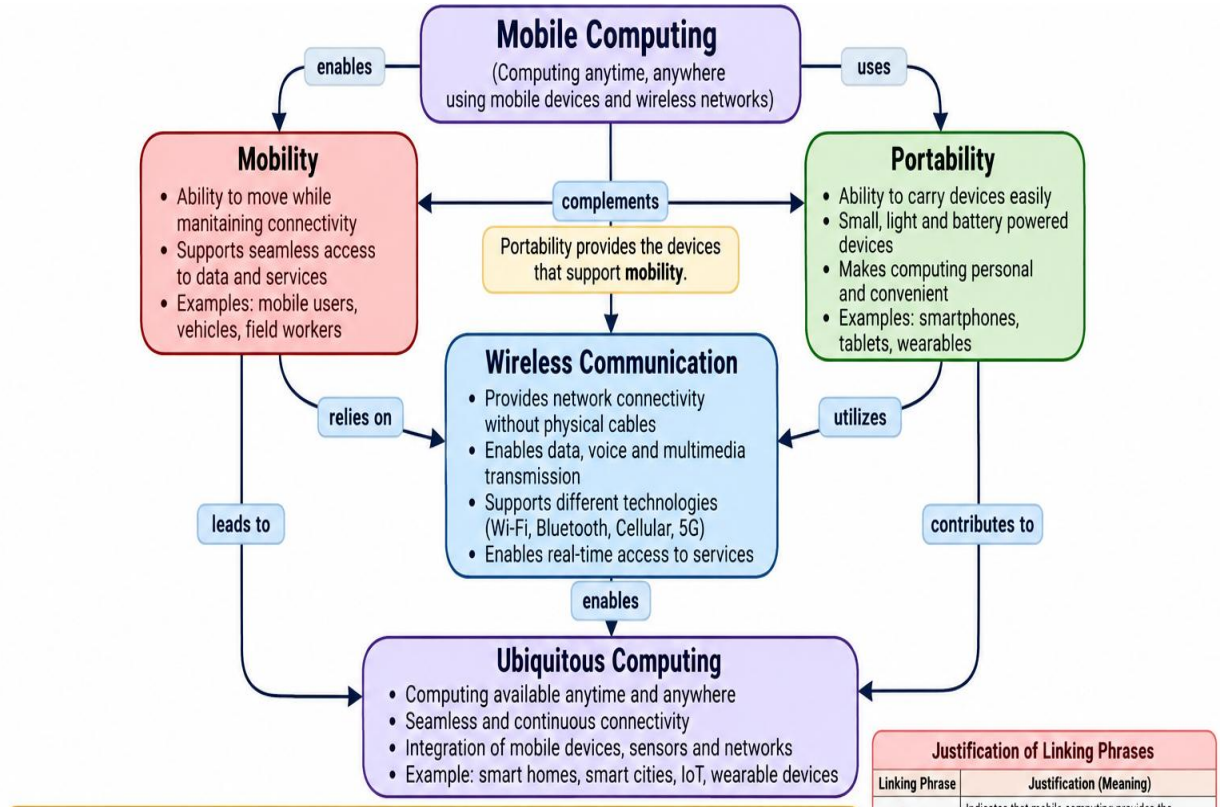
I ___T Bhanu Teja/192472259__ *(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: __Bhanu Teja__

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Annexure A - QUESTIONS

1. Design a concept map that evaluates the significance of the fundamental concepts of mobile computing.
   Establish relationships among mobility, portability, wireless communication, and ubiquitous computing.
   Prioritize the concepts based on their contribution to mobile computing.
   Justify the linking phrases used between the concepts.



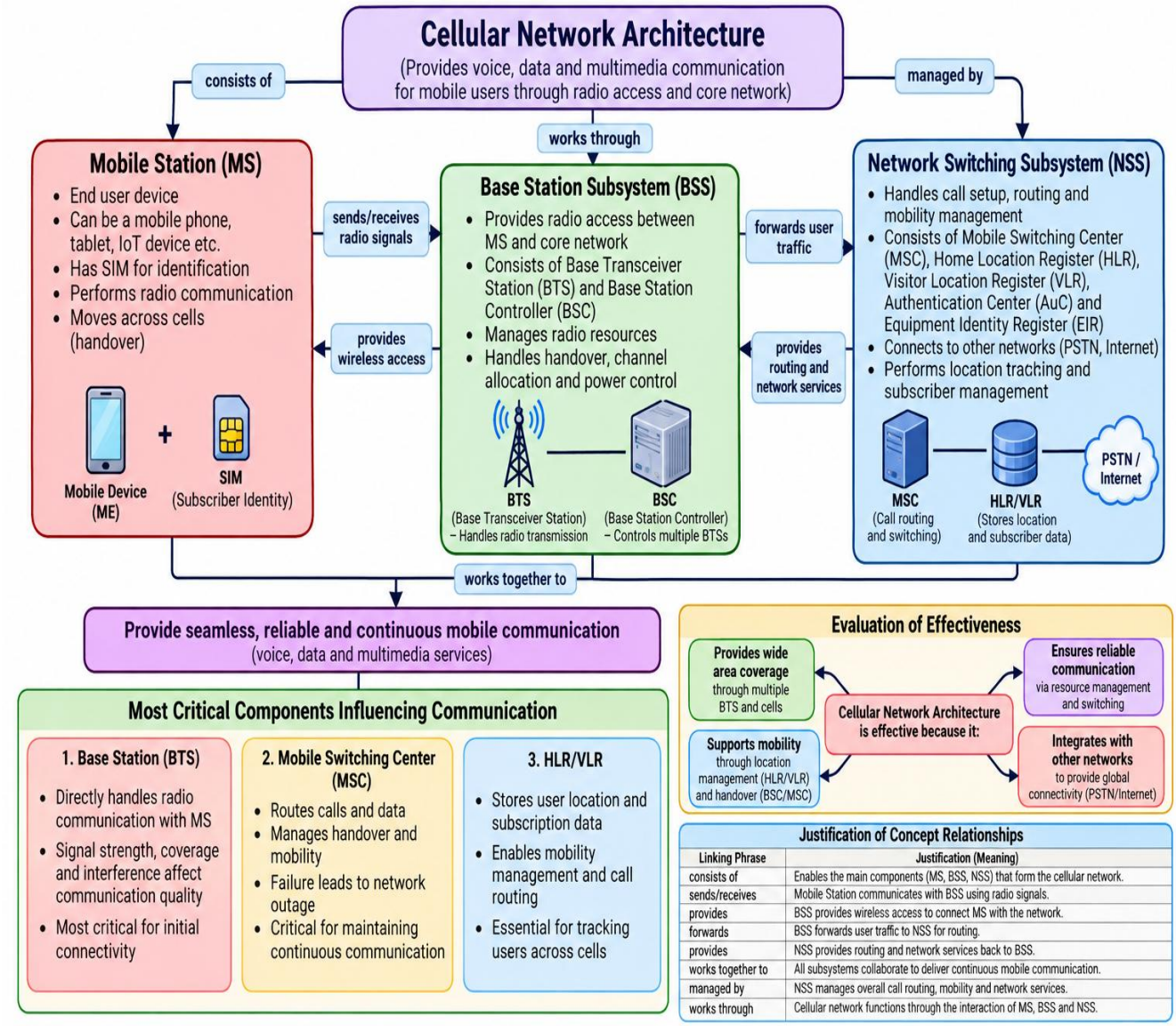
Priority of Concepts (Based on Contribution to Mobile Computing)

1. Wireless Communication	2. Mobility	3. Portability	4. Ubiquitous Computing
• Fundamental enabler of connectivity • Without wireless networks, mobile computing is not possible.	• Core characteristic of mobile computing • Allows users to move while staying connected.	• Provides compact devices for mobility • Makes computing convenient and personal.	• Ultimate goal and extended vision • Combines all the above concepts to provide seamless computing anytime, anywhere.

Justification of Linking Phrases

Linking Phrase	Justification (Meaning)
enables	Indicates that mobile computing provides the capability for mobility and ubiquitous computing.
uses	Shows that mobile computing uses portable devices.
complements	Shows a mutually supportive relationship between mobility and portability.
relies on	Mobility depends on wireless communication for continuous connectivity.
utilizes	Portable devices use wireless communication to access networks.
leads to	Mobility contributes to achieving ubiquitous computing.
contributes to	Portability helps in realizing ubiquitous computing.
enables	Wireless communication makes ubiquitous computing possible.

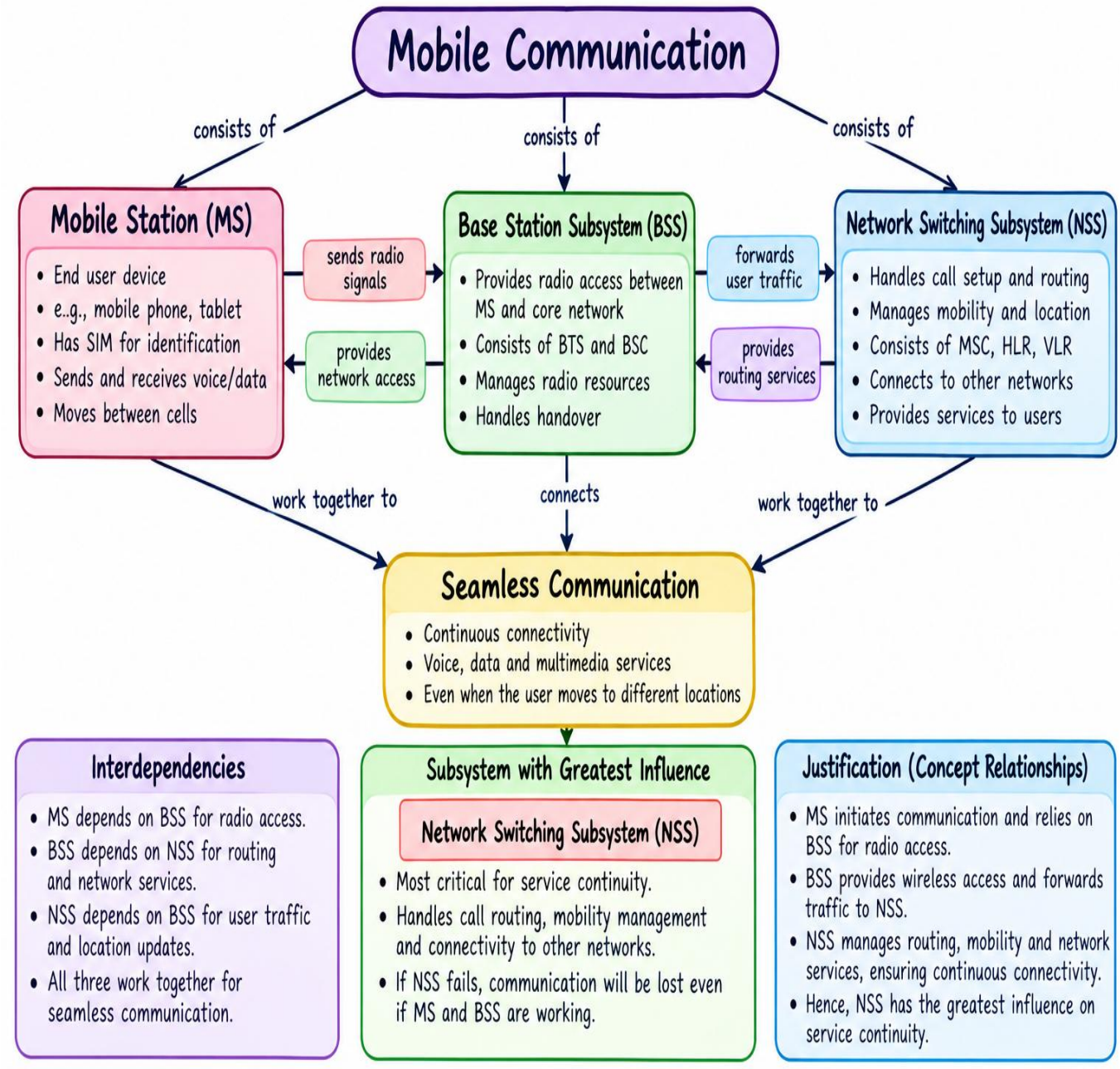
2. Construct a concept map that evaluates the effectiveness of the cellular network architecture.
   Illustrate the interactions among Mobile Station, Base Station Subsystem, and Network Switching Subsystem.
   Identify the most critical components influencing communication.
   Justify your evaluation through meaningful concept relationships.



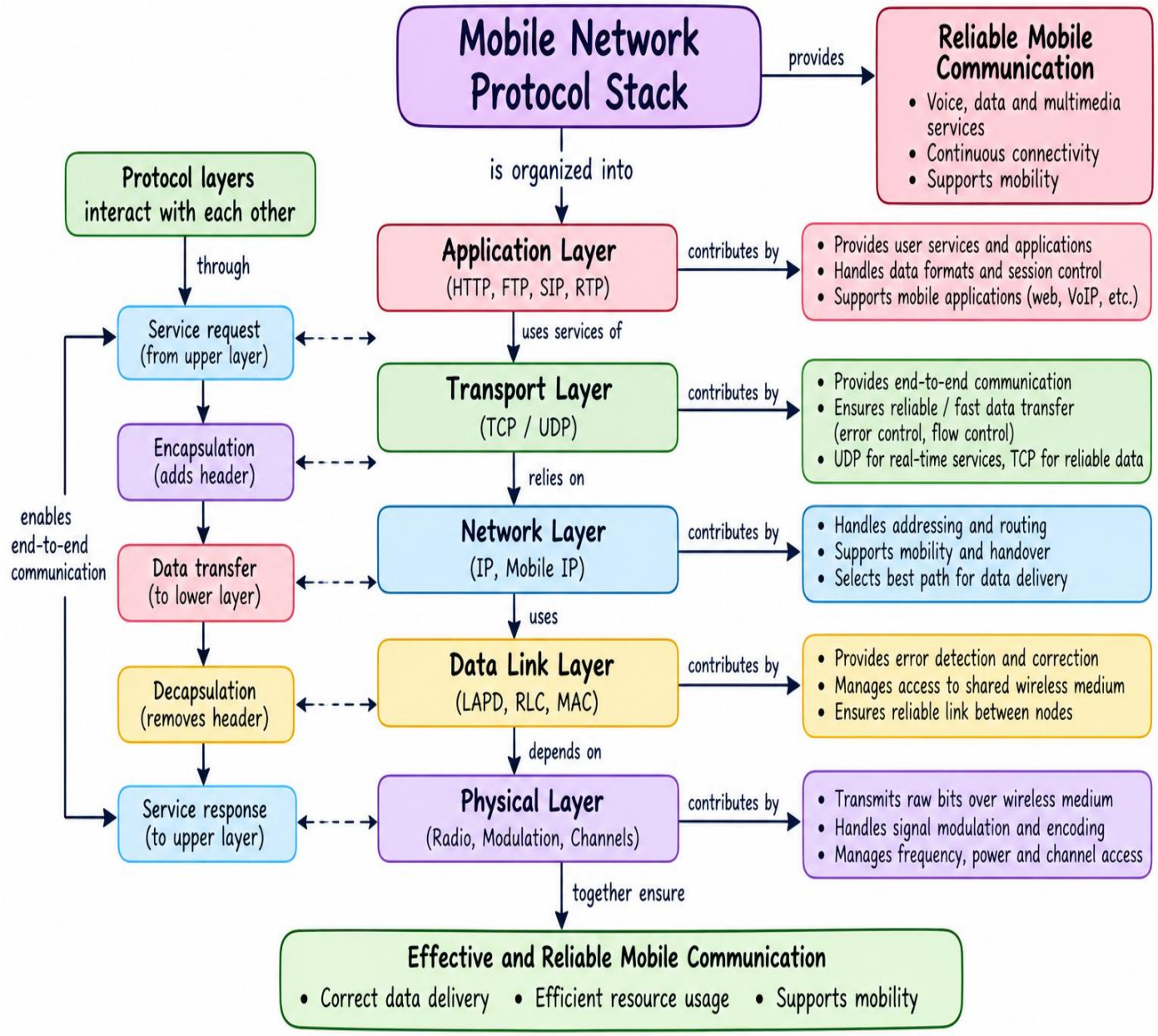
Justification of Concept Relationships

Linking Phrase	Justification (Meaning)
consists of	Enables the main components (MS, BSS, NSS) that form the cellular network.
sends/receives	Mobile Station communicates with BSS using radio signals.
provides	BSS provides wireless access to connect MS with the network.
forwards	BSS forwards user traffic to NSS for routing.
provides	NSS provides routing and network services back to BSS.
works together to	All subsystems collaborate to deliver continuous mobile communication.
managed by	NSS manages overall call routing, mobility and network services.
works through	Cellular network functions through the interaction of MS, BSS and NSS.

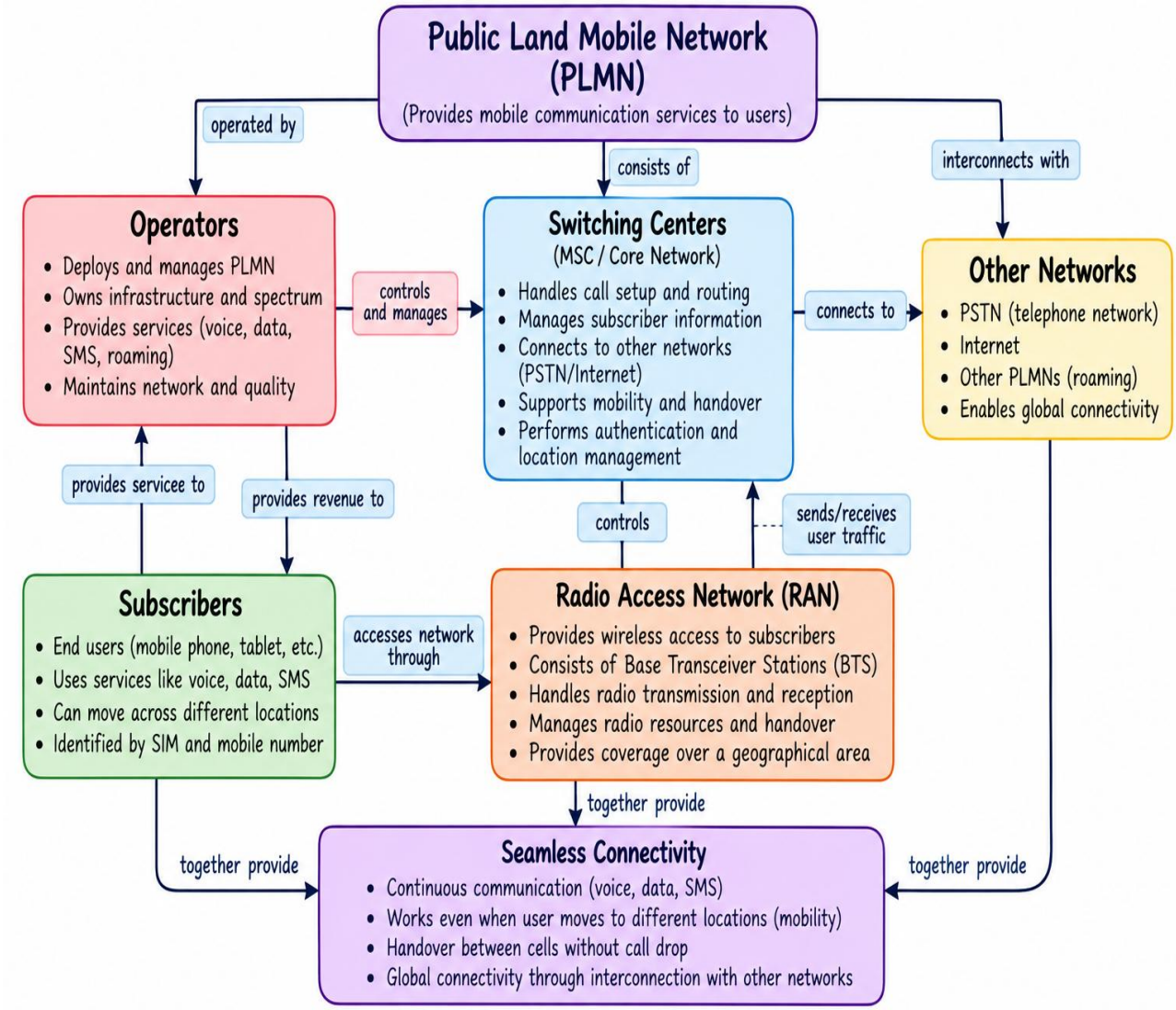
3. Construct a concept map that evaluates the contribution of Mobile Station, Base Station Subsystem, and Network Switching Subsystem toward seamless communication.
   Illustrate their interdependencies.
   Assess which subsystem has the greatest influence on service continuity.
   Justify your conclusions using concept relationships.



4. Develop a concept map that evaluates the effectiveness of the mobile network protocol stack.
   Illustrate interactions among protocol layers.
   Assess how each layer contributes to reliable mobile communication.
   Support your evaluation with meaningful hierarchical relationships.



5. Construct a concept map that evaluates the operational efficiency of a Public Land Mobile Network (PLMN).
   Relate operators, subscribers, switching centers, and radio access networks.
   Assess how these components collectively provide seamless connectivity.
   Justify the relationships included in the concept map.



Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____